Scikit - learn

- Sklearn is an open source ML library that supports supervised & Unsupervised Learning.

→ We as a programmer need not to write the logic / maths behind the algorithms

→ As they are already there in the ML libraries.

→ But we have to know how the algorithms works, the maths behind them, the intuition, the logic, working of the algorithm.

→ As a ML/AI engineer, we have to decide which algo to use where, so that's why we have to understand the working of every algorithm.

→ If you have doubt/ visit the official documentation, and it is a good practice to learn from documentation.

Benefits of Scikit-Learn

1. well-documented
2. easy to learn
3. Works well with numpy & pandas
4. used by professionals as well as beginner's.

Scanned with OKEN Scanner

Algorithms

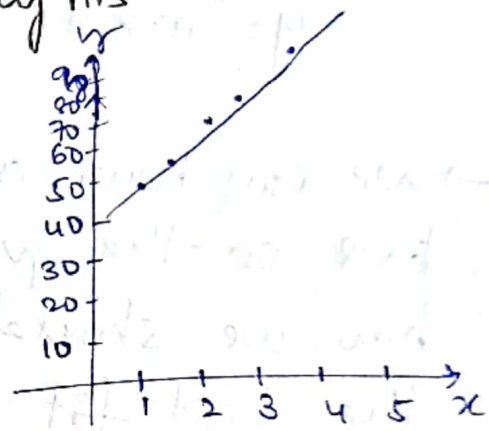
- → Linear Regression → regression type of problem
- → Logistic Regression → classification
- → K-nearest neighbour (KNN) → classification
- → Decision Trees → Both
- → SVM (Support Vector Machine) → Both
- → Naive Bayes → classification
- → Random forest → Both.

Linear Regression

- We have a dataset of study hrs - scores

x (independent) hrs	y (dependent) score (100)
1	50
2	55
3	65
4	75
5	90

again



→ Here the score is continous so this a regression problem.

→ In the above example we want to predict if a student studies for 3.5hrs how many marks he/she will get?

→ We build a linear model train the model evaluate & make new predictions.

Scanned with OKEN Scanner

→ If we want the relationship between 'x' & 'y' in maths we draw a straight line.

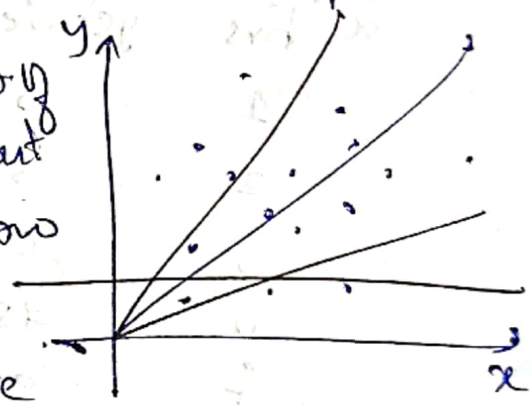
→ Straight line will tell the best relantionship b/t x & 'y'.

<u>Best fit line</u> : Majority points will lie on the line.

Generalized equation of Linear Regression

$$y = mx + c$$

m → slope (coefficient)
c → intercept (bias)



→ We can have any no of lines on the graph; but how we should know the best fit line.

→ We have huge data so we cannot tell by visualizing, if small data we can say that this is best line.

→ Best fit line gives the predicted value for any actual value.

→ The difference b/w the actual & predicted value is known as error. (residual error)

→ Best fit line only predicts correct value for the majority of dataset not all.

Scanned with OKEN Scanner

→ The Bestfit line is that line on the graph which has the minimum error

→ If we have multiple lines on the graph the line which has the minimum error/ residual error is known as the Bestfit line.

Equation of straight line

$$y = c + mx$$

rewrite : $\boxed{h_\theta(x) = \theta_0 + \theta_1 x_1}$

x_1 → 1st feature

$h_\theta(x)$ → hypothesis for x

θ_0 → bias/ intercept

θ_1 → 1st parameter / 1st coefficient

→ In 2D we have only 1 input & 1 output so linear equation gives us a best fit line.

→ In 3D we have 2 features we predict a plane → plane eqn.

→ In 4D → hyperplane eqn

→ We cannot imagine 4D, 5D ... planes, so we are taking example of 2D & finding the Best fit line.

* If we have many features we have eqn.

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 \ldots \ldots \theta_n x_n$$

Scanned with OKEN Scanner

Cost Function

Measures how well the model's predictions match the actual data.

Loss / Cost : Overall error of a model in a dataset is known as Loss.

→ We always try to minimize the loss or Cost function.

For ith point (any point)

Pred-value − actual-value

$$h_\theta(x^{(i)}) - y^{(i)}$$

→ In Cost function we take the error of every value & sum it.

$$\sum_{i=1}^{m} h_\theta(x^{(i)}) - y^{(i)} \quad \text{for m-datapoints}$$

→ There is one cache in it, for some values there will +ve point/error for some there will be -ve error & then the values will be cancelled. (upper -ve, below +ve) from predicted

→ To avoid this we use multiple different cost functions to calculate the error.

→ Squared error

→ log of error

→ absolute error

Scanned with OKEN Scanner

→ For Linear Regression typically we use MSE (Mean squared error)

$$J(\theta) = \frac{1}{m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)} \right)^2$$

→ We calculate $J(\theta)$ for every line & the line whose $J(\theta)$ is least we use it as Best Fit line.

→ In ML we use $\frac{1}{2m}$ in $J(\theta)$, finally our cost function for Linear Regression is

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)} \right)^2$$

** Remember the eqn for best fit line & cost function

→ We know Both, but how do we actually solve these & know where our best fit line appears.

→ We need to find the correct value of θ_0, θ_1 in Best fit line, that is our ultimate goal.

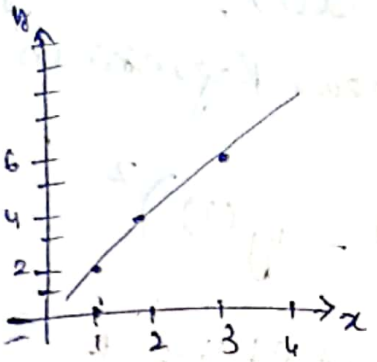
→ To do these we follow simple steps

- Initial the θ_0, θ_1 with random values
- Calculate the $J(\theta)$ for the values initialized

Scanned with OKEN Scanner

- check the $J(\theta)$ if it can be minimized more
- Update the values of θ_0, θ_1
- Again check the $J(\theta)$, repeat these until we find the best value for $J(\theta)$.

<u>Example</u> : datapoints = (1,2) (2,4) (3,6)



$$h_\theta(x) = \theta_0 + \theta_1 x_1$$

$$J(\theta) = \frac{1}{2.m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)}\right)^2$$

<u>Case1</u> : $\theta_0 = 0$, $\theta_1 = 2$ } random values

$$h_\theta(x) = 2x$$

$$J(\theta) = \frac{1}{6}\left(0^2 + 0^2 + 0^2\right) = 0$$

$$\theta_1 = 2 \; ; \; J(\theta) = 0$$

$2(1) = 2$ → output (input)
$2(2) = 4$
$2(3) = 6$

<u>Case2</u> : $\theta_0 = 0$, $\theta_1 = 1 \Rightarrow h_\theta(x) = x$

$$J(\theta) = \frac{1}{2 \cdot 6}\left(1^2 + 2^2 + 3^2\right) = \frac{14}{6} = 2.34$$

$$\theta_1 = 1 \; , \; J(\theta) = 2.34$$

$1 - 2 = 1$
$2 - 4 = 2$
$3 - 6 = 3$

Scanned with OKEN Scanner

Case 3 : $\theta_0 = 0$, $\theta_1 = 3$

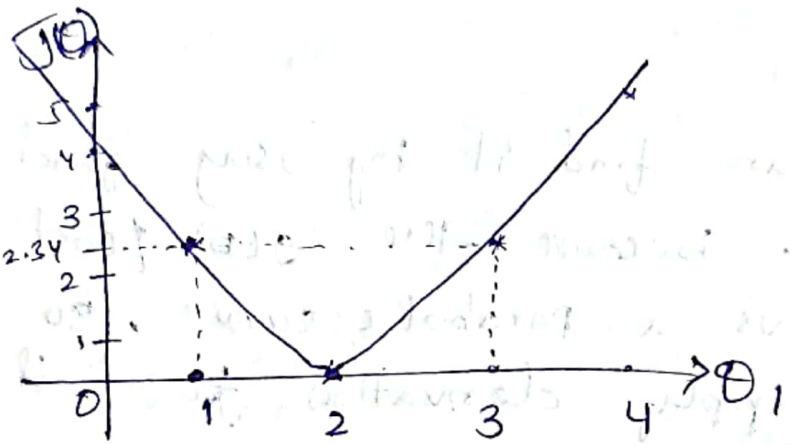
$h_\theta(x) = 3x$

$J(\theta) = \frac{1}{6}\left(1^2 + 2^2 + 3^2\right)$

$\theta_1 = 3$, $J(\theta) = 2.34$

$3(1) = 3 - 2 = 1$
$3(2) = 6 - 4 = 2$
$3(3) = 9 - 6 = 3$

Plotting the θ_1 values & $J(\theta)$ on the graph we get.



→ This curve basically represents the Cost function

→ If we increase theta (θ) or decrease the (θ) the cost function will also increase.

→ There will be a θ_1 where the cost function or $J(\theta)$ value will be minimum.

→ and the point is known as global minimum

→ The curve signifies the value of the cost function

Scanned with OKEN Scanner

→ and we try to find the value of global minima.

→ The process of finding the global minima is known as <u>Gradient Descent</u>.

→ We need not to take random values of theta (θ_1) to find the global minima.

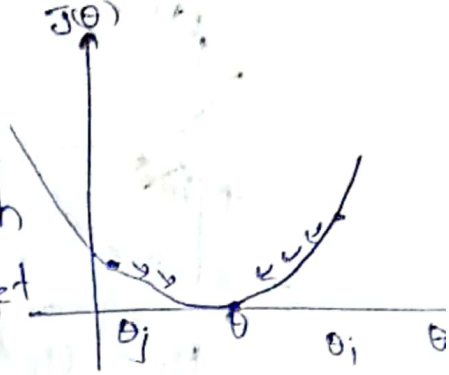
→ We can find it by using gradient descent. because the $J(\theta)$ / cost function gives us a parabolic curve, so we can apply derivation to find the global minima.

<u>Gradient Descent</u>

It is an iterative optimization algorithm used to minimize a cost function by adjusting model parameters in the direction of the steepest descent of the function's gradient.

$\frac{\partial(J(\theta))}{\partial\theta}$	<u>Gradient</u>	<u>Descent</u>
	↓	↓
	slope	moving downward

→ α → learning Rate



→ GD says we start from any point, the algo say to which direction we have move to get to the global minima.

→ The small steps we take to reach the global minima is known as learing rate.

→ In derivatives we can directly find the global minima, In ML we cannot because the derivates gives a complex value after solving the GD.

→ We use convergence theorem to find the global minima in the curve using derivatives.

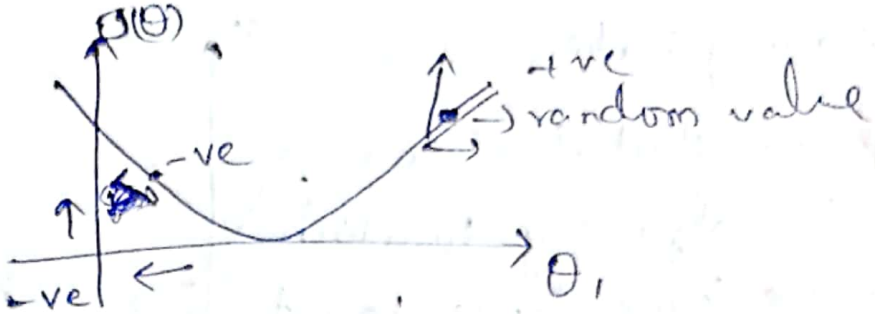
→ This is not done in a single go, it is an iterative process.

→ Convergence theorem tells us how much we need to make change in our (θ) theta.

$$\theta_K = \theta_K - \alpha . \frac{\partial (J(\theta))}{\partial \theta_K}$$

→ Suppose we start from a random theta (θ) in a given linear problem

Scanned with OKEN Scanner



$$\theta_1 = \theta_1 - \alpha \frac{\partial(J(\theta))}{\partial \theta_1}$$

→ If the value of G.D is +ve we are on the right of the curve

→ In G.D we find the rate of change with respective 'x', how much our y changes if we change x.

• $CT = \theta_1 = \theta_1 - \alpha \overset{+ve}{(\text{slope})}$

→ if the value is on the right side we get the slope value as +ve so we substract the slope & learning rate from the θ_1.

→ after substracting the θ_1 value gets decreased so we move on the left side of the curve (slightly).

→ $\frac{1}{2.m}$ hum log isliye lete cost func me kiyuki jab baad mein derivative apply karte tho wo square se cancel hojata. easy calculation

Scanned with OKEN Scanner

→ if α → too small → slow training
α → too large → divergence, miss minima.

→ we try to take the appropriate value of α - (learning rate).

<u>Steps we follow</u>

① θ_0, θ_1 - initialize with random values
② make predictions + calculate $J(\theta)$
③ use Convergence theorem to update the parameters
④ Repeat until we get global minima.

Scanned with OKEN Scanner